

Adopted Levels, Gammas

$Q(\beta^-) = -11874.4$ 9; $S(n) = 12740.3$ 7; $S(p) = 5896.2$ 7; $Q(\alpha) = -6429.7$ 7 (2021Wa16)
 $Q(\beta^-n) = -29632$ 17, from mass excesses of -1487 17 for ^{34}K measured by (2024Dr01); -23047.3 7 for ^{35}Ar from (2021Wa16). Value from (2021Wa16): $Q(\beta^-n) = -29900$ 200 (syst).
 $S(2n) = 29805.6$ 8, $S(2p) = 11039.4$ 7, $Q(\epsilon) = 5966.2$ 7 (2021Wa16).
 Isotope discovery (2012Th10): $^{32}\text{S}(\alpha,n)^{35}\text{Ar}$ at Purdue (1940Ki12,1941Ki01,1941El04).
 ^{35}Ar production:
 2012Zh06: $^9\text{Be}, ^{181}\text{Ta}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar}) = 57$ MeV/nucleon at HIRFL. Measured momentum distributions and production cross sections of fragments. Observed competition between projectile fragmentation and other mechanisms. Compared with EPAX, abrasion- ablation, and HIPSE models. Studied target dependence of fragment cross sections.
 2007No13: $^{181}\text{Ta}(^{40}\text{Ar}, X)$ at $E(^{40}\text{Ar}) = 100$ MeV/nucleon at RIKEN. Measured fragment momentum distributions and production cross sections.
 ^{35}Ar decay measurements:
 ^{35}Ar produced via $^{35}\text{Cl}(p,n)^{35}\text{Ar}$ reaction (1985Da04,1984Ad01,1980Wi13,1979Ha04,1977Az01,1971Ge04,1971De05,1969Wi18,1969Fr09,1960Wa04,1956Ki29). Measured $\epsilon + \beta^+$ -delayed γ rays, decay branching ratios, and parent ^{35}Ar $T_{1/2}$.
 ^{35}Ar radius measurements:
 2002Oz03: $\text{C}(^{35}\text{Ar}, X)$ at $E(^{35}\text{Ar}) \approx 950$ MeV/nucleon at RIKEN. Measured interaction cross sections. Deduced effective radii and proton skin features.
 2000Ge20: ^{35}Ar produced at ISOLDE. Measured β asymmetry and hyperfine structure using β -NMR spectroscopy. Deduced mean squared charge radii and quadrupole moments.
 1996Ki04,1995KiZZ: ^{35}Ar produced by ISOLDE. Measured isotope shifts and hyperfine structure using collinear fast-beam laser spectroscopy. Deduced mean square charge radii and electric quadrupole moments.
 ^{35}Ar mass measurement: (2011Tu09).
 Theoretical calculations: (2020Ri06, 2020RiZX, 2020RiZZ).

 ^{35}Ar LevelsCross Reference (XREF) Flags

A	^{35}K ϵ decay (175 ms)	E	$^{24}\text{Mg}(^{16}\text{O}, \alpha n \gamma)$	I	$^{36}\text{Ar}(p, d)$
B	^{36}Ca ϵp decay (100.9 ms)	F	$^{32}\text{S}(\alpha, n)$	J	$^{36}\text{Ar}(d, t)$
C	$^1\text{H}(^{36}\text{Ar}, d)$	G	$^{33}\text{S}(^3\text{He}, n \gamma)$	K	$^{36}\text{Ar}(^3\text{He}, \alpha)$
D	$^{16}\text{O}(^{24}\text{Mg}, \alpha n \gamma)$	H	$^{35}\text{Cl}(^3\text{He}, t)$		

$E(\text{level})^\dagger$	J^π	$T_{1/2}$	XREF	Comments
0.0	$3/2^+$	1.7755 s 14	ABCDEFG IJK	$\% \epsilon + \% \beta^+ = 100$ $\mu = +0.6322$ 2 (2002Ma41,2019StZV) $Q = -0.084$ 15 (1996Ki04,2021StZZ) μ : β -NMR (2002Ma41). Others: $+0.632$ 2 (1965Ca04), $+0.633$ 7 (1996Ki04) using β -NMR. Q : β -NMR (1996Ki04). J^π : $L(p, d) = L(d, t) = L(^3\text{He}, \alpha) = L(^1\text{H}(^{36}\text{Ar}, d) = 2$ from 0^+ . Allowed $\epsilon + \beta^+$ feedings to $1/2^+$ levels in ^{35}Cl . Mirror level: $3/2^+$ ^{35}Cl g.s. $T_{1/2}$: weighted average of 1.83 s 3 (1956Ki29), 1.83 s 2 (1959Al10), 1.79 s 1 (1960Ja12), 1.84 s 10 (1960Wa04), 1.76 s 3 (1963Ne05), 1.770 s 6 (1969Wi18), 1.787 s 12 (1971Ge04), 1.774 s 3 (1977Az01), and 1.7754 s 11 (2006Ia05). Evaluated rms nuclear charge radius $R = 3.3636$ fm 42 (2013An02). XREF: F(890) $E(\text{level})$: 1963Ne05 (α, n) observed the first excited state in ^{35}Ar at 890 50 keV. J^π : $L(p, d) = L(d, t) = L(^3\text{He}, \alpha) = 0$ from 0^+ . XREF: F(2030)I(1700)J(1700)
1184.08 25	$1/2^+$		ABC FG IJK	
1750.78 22	$(5/2)^+$		A DEFG IJK	

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Adopted Levels, Gammas (continued) ^{35}Ar Levels (continued)

E(level) [†]	J ^π	XREF	Comments
			XREF: F(2030)I(1700)J(1700)
			E(level): 1963Ne05 (α, n) observed the second excited state in ^{35}Ar at 2030 80 keV.
2603.22 28	7/2 ⁽⁺⁾	DE G	J ^π : L($^3\text{He}, \alpha$)=2 from 0 ⁺ . Mirror level: 5/2 ⁺ at 1763 keV in ^{35}Cl .
2638.01 26	3/2 ⁺	A IJK	J ^π : $\Delta J=2$ γ to 3/2 ⁺ in ($^{16}\text{O}, \alpha n \gamma$). Mirror level: 7/2 ⁺ at 2646 keV in ^{35}Cl .
			XREF: I(2615)K(2649)
2982.79 12	5/2 ⁺	A C IJK	J ^π : L(p,d)=L($^3\text{He}, \alpha$)=2 from 0 ⁺ with J dependence in (p,d).
			XREF: I(2970)
3196.98 [‡] 26	7/2 ⁻	CDE G IJK	J ^π : L(p,d)=L(d,t)=L($^3\text{He}, \alpha$)=2 from 0 ⁺ with J dependence in (p,d).
			J ^π : L(p,d)=L($^3\text{He}, \alpha$)=3 from 0 ⁺ . $\Delta J=1$ γ to (5/2) ⁺ and $\Delta J=2$ γ to 3/2 ⁺ in ($^{16}\text{O}, \alpha n \gamma$) and ($^{24}\text{Mg}, \alpha n \gamma$); band assignment.
3882 5	1/2 ⁺	K	J ^π : L($^3\text{He}, \alpha$)=0 from 0 ⁺ .
4001 3	1/2 ⁻ , 3/2 ⁻	K	J ^π : L($^3\text{He}, \alpha$)=1 from 0 ⁺ .
4065.0? 4	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	A	XREF: A(?)
			J ^π : possibly allowed $\varepsilon + \beta^+$ feeding from 3/2 ⁺ parent with log ft=5.6 +4-2.
4113 4		K	
4135 4	1/2 ⁻ , 3/2 ⁻	K	J ^π : L($^3\text{He}, \alpha$)=1 from 0 ⁺ .
4359.0 5	(9/2 ⁻)	DE K	J ^π : $\Delta J=(1)$ γ to 7/2 ⁻ in ($^{16}\text{O}, \alpha n \gamma$). Possible mirror level: 9/2 ⁻ at 4348 keV in ^{35}Cl .
4528.3 4	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺)	A K	XREF: K(4515)
			J ^π : possibly allowed $\varepsilon + \beta^+$ feeding from 3/2 ⁺ parent with log ft=5.4 +4-2.
4725.9 6	1/2 ⁺	A Hi K	XREF: i(4756)K(4713)
			J ^π : L($^3\text{He}, \alpha$)=0 from 0 ⁺ . Other: L(p,d)=0 from 0 ⁺ for a group at 4756 28.
4785.8 11	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	A Hi K	XREF: i(4756)K(4774)
			J ^π : allowed $\varepsilon + \beta^+$ feeding from 3/2 ⁺ parent with log ft=5.2 2. Other: L(p,d)=0 from 0 ⁺ for a group at 4756 28.
5059 11		K	
5116 2	3/2 ⁺ , 5/2 ⁺	HI K	E(level): weighted average of 5102 20 from (p,d) and 5116 2 from ($^3\text{He}, \alpha$).
			J ^π : L($^3\text{He}, \alpha$)=2 from 0 ⁺ in 1973Be26 . Discrepancy: L(p,d)=3 from 0 ⁺ (1968Ko04). 1973Be26 also considered L=3 but found L=2 gives a much better fit of the data than L=3 does.
5207 3		H K	E(level): from ($^3\text{He}, \alpha$).
5384.2 [‡] 4	(11/2 ⁻)	DE HI K	XREF: I(5400)
			J ^π : $\Delta J=2$ γ to 7/2 ⁻ in ($^{16}\text{O}, \alpha n \gamma$) and ($^{24}\text{Mg}, \alpha n \gamma$); band assignment. γ to (9/2 ⁻) in ($^{16}\text{O}, \alpha n \gamma$) and ($^{24}\text{Mg}, \alpha n \gamma$). Possible mirror level: 11/2 ⁻ at 5407 keV in ^{35}Cl .
5482 2	3/2 ⁺ , 5/2 ⁺	H K	E(level): from ($^3\text{He}, \alpha$).
			J ^π : L($^3\text{He}, \alpha$)=2 from 0 ⁺ .
5572.67 15	3/2 ⁺	A G	T=3/2
			XREF: G(5537)
			J ^π : isobaric analog state of 3/2 ⁺ ^{35}K g.s. with log ft=3.31 4. L($^3\text{He}, n$)=(0) from 3/2 ⁺ . $^{35}\text{Cl}(^3\text{He}, t)$ attempted to search this T=3/2 level, but did not find it.
5594 2	3/2 ⁺ , 5/2 ⁺	C HI K	XREF: C(5570)
			E(level): weighted average of 5598 20 from (p,d) and 5594 2 from ($^3\text{He}, \alpha$).
			J ^π : L(p,d)=L($^3\text{He}, \alpha$)=2 from 0 ⁺ .
			Evaluators consider the 5994 level to be different from the 5572.67 T=3/2 level because (p,d) and ($^3\text{He}, \alpha$) from T=0 targets should not populate T=3/2 levels.
5613.6 9	(11/2 ⁻)	E	J ^π : Possible mirror level: 11/2 ⁻ at 5927 keV in ^{35}Cl .
5765.8 5	(13/2 ⁻)	DE	J ^π : $\Delta J=1$ γ to (11/2 ⁻) in ($^{16}\text{O}, \alpha n \gamma$) and ($^{24}\text{Mg}, \alpha n \gamma$). $\Delta J=(2)$ γ to (9/2 ⁻) in ($^{16}\text{O}, \alpha n \gamma$). Possible mirror level: 13/2 ⁻ at 6087 keV in ^{35}Cl .

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Adopted Levels, Gammas (continued) ^{35}Ar Levels (continued)

E(level) [†]	J ^π	XREF	Comments
5915 3		H JK	E(level): weighted average of 5913 5 from (d,t) and 5916 3 from ($^3\text{He},\alpha$).
5991 3		J	
6037 3	3/2 ⁺ , 5/2 ⁺	HIJK	XREF: I(6024) E(level): weighted average of 6037 3 from (d,t) and 6036 3 from ($^3\text{He},\alpha$). Other: 6024 20 from (p,d). J ^π : L(p,d)=L($^3\text{He},\alpha$)=2 from 0 ⁺ . XREF: J(?)
6055? 3		J	
6076 3		J	
6163 2		JK	E(level): weighted average of 6164 3 from (d,t) and 6162 2 from ($^3\text{He},\alpha$).
6253 3		Jk	XREF: k(6262)
6273 3		Jk	XREF: k(6262)
6302 3		J	
6332 3		J	
6345 3	(1/2, 3/2, 5/2)	A J	E(level): From (d,t). Other: 6348 11 from ^{35}K ε decay. J ^π : $\varepsilon+\beta^+$ feeding from 3/2 ⁺ parent with log ft=7.2 1.
6415 2		J	
6439? 4		J	XREF: J(?)
6460 3		J	
6523 3		J	
6557 3		J	
6585 3		J	
6606 3		iJ	XREF: i(6620)
6616 2	1/2 ⁺	iJK	XREF: i(6620) E(level): weighted average of 6617 2 from (d,t) and 6615 3 from ($^3\text{He},\alpha$). Other: 6620 30 from (p,d). J ^π : L($^3\text{He},\alpha$)=L(p,d)=0 from 0 ⁺ . XREF: i(6620)
6644 3		iJ	XREF: i(6620)
6651 3		iJ	XREF: i(6620)
6673 4	5/2 ⁻ , 7/2 ⁻	IJ	XREF: I(6700) E(level): weighted average of 6700 20 from (p,d) and 6672 3 from (d,t). J ^π : L(p,d)=3 from 0 ⁺ .
6823 2	3/2 ⁺ , 5/2 ⁺	I K	E(level): from ($^3\text{He},\alpha$). Other: 6820 30 from (p,d). J ^π : L(p,d)=2 from 0 ⁺ .
6948 2		K	
7044 4	3/2 ⁺ , 5/2 ⁺	A I K	XREF: I(7030) E(level): weighted average of 7053 11 from ^{35}K ε decay, 7030 20 from (p,d), and 7043 4 from ($^3\text{He},\alpha$). J ^π : L(p,d)=2 from 0 ⁺ .
7117 10		K	
7255 11		A	
7289 10		A K	E(level): weighted average of 7283 11 from ^{35}K ε decay and 7293 10 from ($^3\text{He},\alpha$).
7427 10		A K	E(level): weighted average of 7431 11 from ^{35}K ε decay and 7423 10 from ($^3\text{He},\alpha$).
7509 10	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	A K	E(level): weighted average of 7518 11 from ^{35}K ε decay and 7502 10 from ($^3\text{He},\alpha$). J ^π : allowed $\varepsilon+\beta^+$ feeding from 3/2 ⁺ parent with log ft<5.0.
7840 10		K	
8019 10		K	
8109.7 [±] 13	(15/2 ⁻)	E	J ^π : γ to (11/2 ⁻) and (13/2 ⁻) in ($^{16}\text{O},\alpha\gamma$); band assignment. Possible mirror level: 15/2 ⁻ at 8319 keV in ^{35}Cl .
8212.6 8	(15/2 ⁻)	E	J ^π : $\Delta J=2$ γ to (11/2 ⁻) and γ to (13/2 ⁻) in ($^{16}\text{O},\alpha\gamma$). Possible mirror level: 15/2 ⁻ at 8487 keV in ^{35}Cl .
8393? 20	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺	A	XREF: A(?)

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Adopted Levels, Gammas (continued) ^{35}Ar Levels (continued)

<u>E(level)[†]</u>	<u>J^π</u>	<u>XREF</u>	<u>Comments</u>
			XREF: A(?) E(level): From (^{35}K ε decay). J ^π : allowed $\varepsilon+\beta^+$ feeding from 3/2 ⁺ parent with log ft=4.6 +3-2.
9906.0 [‡] 20	(19/2 ⁻)	E	J ^π : $\Delta J=2$ γ to (15/2 ⁻) in ($^{16}\text{O},\alpha\gamma$); band assignment. Possible mirror level: 19/2 ⁻ at 10180 keV in ^{35}Cl .
12277.0 [‡] 32	(23/2 ⁻)	E	J ^π : $\Delta J=2$ γ to (19/2 ⁻) in ($^{16}\text{O},\alpha\gamma$); band assignment. Possible mirror level: 23/2 ⁻ at 12571 keV in ^{35}Cl .

[†] E(level) from a least-squares fit to γ -ray energies for levels connected with γ transitions; from particle-transfer reactions or ^{35}K $\varepsilon+\beta^+$ -delayed proton decays for other levels.

[‡] Band(A): Band based on f_{7/2} orbital.

<u>$\gamma(^{35}\text{Ar})$</u>							<u>Comments</u>
<u>E_i(level)</u>	<u>J_i^π</u>	<u>E_γ[†]</u>	<u>I_γ[†]</u>	<u>E_f</u>	<u>J_f^π</u>	<u>Mult.[‡]</u>	
1184.08	1/2 ⁺	1184.1 3	100	0.0	3/2 ⁺		E _γ : weighted average of 1184.0 3 from ^{35}K ε decay and 1184.3 4 from ^{36}Ca εp decay.
1750.78	(5/2) ⁺	1750.6 3	100	0.0	3/2 ⁺		E _γ : weighted average of 1750.5 3 from ^{35}K ε decay, 1750.7 4 from ($^{24}\text{Mg},\alpha\gamma$), and 1750.8 5 from ($^{16}\text{O},\alpha\gamma$).
2603.22	7/2 ⁽⁺⁾	851.9 9	12.3 33	1750.78	(5/2) ⁺		E _γ : weighted average of 852 1 from ($^{24}\text{Mg},\alpha\gamma$) and 851.8 9 from ($^{16}\text{O},\alpha\gamma$).
		2603.0 5	100 10	0.0	3/2 ⁺	Q	I _γ : weighted average of 10 5 from ($^{24}\text{Mg},\alpha\gamma$) and 13.3 33 from ($^{16}\text{O},\alpha\gamma$). E _γ : weighted average of 2603.0 5 from ($^{24}\text{Mg},\alpha\gamma$) and 2602.6 15 from ($^{16}\text{O},\alpha\gamma$). I _γ : other: 100 22 from ($^{24}\text{Mg},\alpha\gamma$).
2638.01	3/2 ⁺	886.8 [#] 5	16 [#] 6	1750.78	(5/2) ⁺		
		2638.0 [#] 4	100 [#] 13	0.0	3/2 ⁺		
2982.79	5/2 ⁺	1798.9 [#] 5	3.5 [#] 6	1184.08	1/2 ⁺		
		2982.68 [#] 13	100 [#] 4	0.0	3/2 ⁺		
3196.98	7/2 ⁻	593.7 2	16.4 30	2603.22	7/2 ⁽⁺⁾		E _γ : weighted average of 593 1 from ($^{24}\text{Mg},\alpha\gamma$) and 593.7 2 from ($^{16}\text{O},\alpha\gamma$).
		1446.2 2	100 8	1750.78	(5/2) ⁺	D	I _γ : weighted average of 16 8 from ($^{24}\text{Mg},\alpha\gamma$) and 16.4 30 from ($^{16}\text{O},\alpha\gamma$). E _γ : weighted average of 1446.2 2 from ($^{24}\text{Mg},\alpha\gamma$), 1446.1 6 from ($^{16}\text{O},\alpha\gamma$), and 1446.0 6 from ($^3\text{He},n\gamma$).
		3197.0 7	21 5	0.0	3/2 ⁺	Q	I _γ : other: 100 9 from ($^{24}\text{Mg},\alpha\gamma$). E _γ : from ($^{24}\text{Mg},\alpha\gamma$). Other: 3197 6 from ($^{16}\text{O},\alpha\gamma$). I _γ : weighted average of 18 5 from ($^{24}\text{Mg},\alpha\gamma$) and 24 5 from ($^{16}\text{O},\alpha\gamma$).

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Adopted Levels, Gammas (continued)

$\gamma(^{35}\text{Ar})$ (continued)						
$E_i(\text{level})$	J_i^π	$E_\gamma^\dagger$	$I_\gamma^\dagger$	E_f	J_f^π	Mult. [‡]
4065.0? 4359.0	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺) (9/2 ⁻)	1426.8 [#] 4 1162.0 8	100 65 24	2638.01 3196.98	3/2 ⁺ 7/2 ⁻	(D)
		1756 1	100 15	2603.22	7/2 ⁽⁺⁾	
4528.3 4725.9	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺) 1/2 ⁺	4527.9 [#] 7 3542.0 [#] 6	100 100 [#] 21	0.0 1184.08	3/2 ⁺ 1/2 ⁺	
		4724.5 [#] 11	41 [#] 17	0.0	3/2 ⁺	
4785.8 5384.2	1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺ (11/2 ⁻)	4785.4 [#] 11 1025.2 4	100 14 4	0.0 4359.0	3/2 ⁺ (9/2 ⁻)	
		2187.1 4	100 6	3196.98	7/2 ⁻	Q
5572.67	3/2 ⁺	1044.4 [#] 4 1507.4 [#] 5 2589.8 [#] 1 2934.5 [#] 5 3821.7 [#] 7 4387.2 [#] 9 5572.3 [#] 10	2.5 [#] 8 3.7 [#] 8 100 [#] 4 6.7 [#] 12 6.7 [#] 14 6.7 [#] 16 11.7 [#] 31	4528.3 4065.0? 2982.79 2638.01 1750.78 1184.08 0.0	(1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺) (1/2 ⁺ , 3/2 ⁺ , 5/2 ⁺) 5/2 ⁺ 3/2 ⁺ (5/2 ⁺) 1/2 ⁺ 3/2 ⁺	
5613.6 5765.8	(11/2 ⁻) (13/2 ⁻)	1254.6 8 381.6 1	100 100 10	4359.0 5384.2	(9/2 ⁻) (11/2 ⁻)	D
		1406.9 7	17.2 35	4359.0	(9/2 ⁻)	(Q)
8109.7	(15/2 ⁻)	2342.6 28 2725.7 14	100 25 50 13	5765.8 5384.2	(13/2 ⁻) (11/2 ⁻)	
8212.6	(15/2 ⁻)	2446.6 16	21 7	5765.8	(13/2 ⁻)	
		2828.3 7	100 18	5384.2	(11/2 ⁻)	Q
9906.0	(19/2 ⁻)	1693.3 27	100 20	8212.6	(15/2 ⁻)	Q
		1796.3 25	67 20	8109.7	(15/2 ⁻)	Q
12277.0	(23/2 ⁻)	2370.9 25	100	9906.0	(19/2 ⁻)	Q

[†] From ($^{16}\text{O}, \alpha\gamma$), unless otherwise noted.

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Adopted Levels, Gammas (continued)

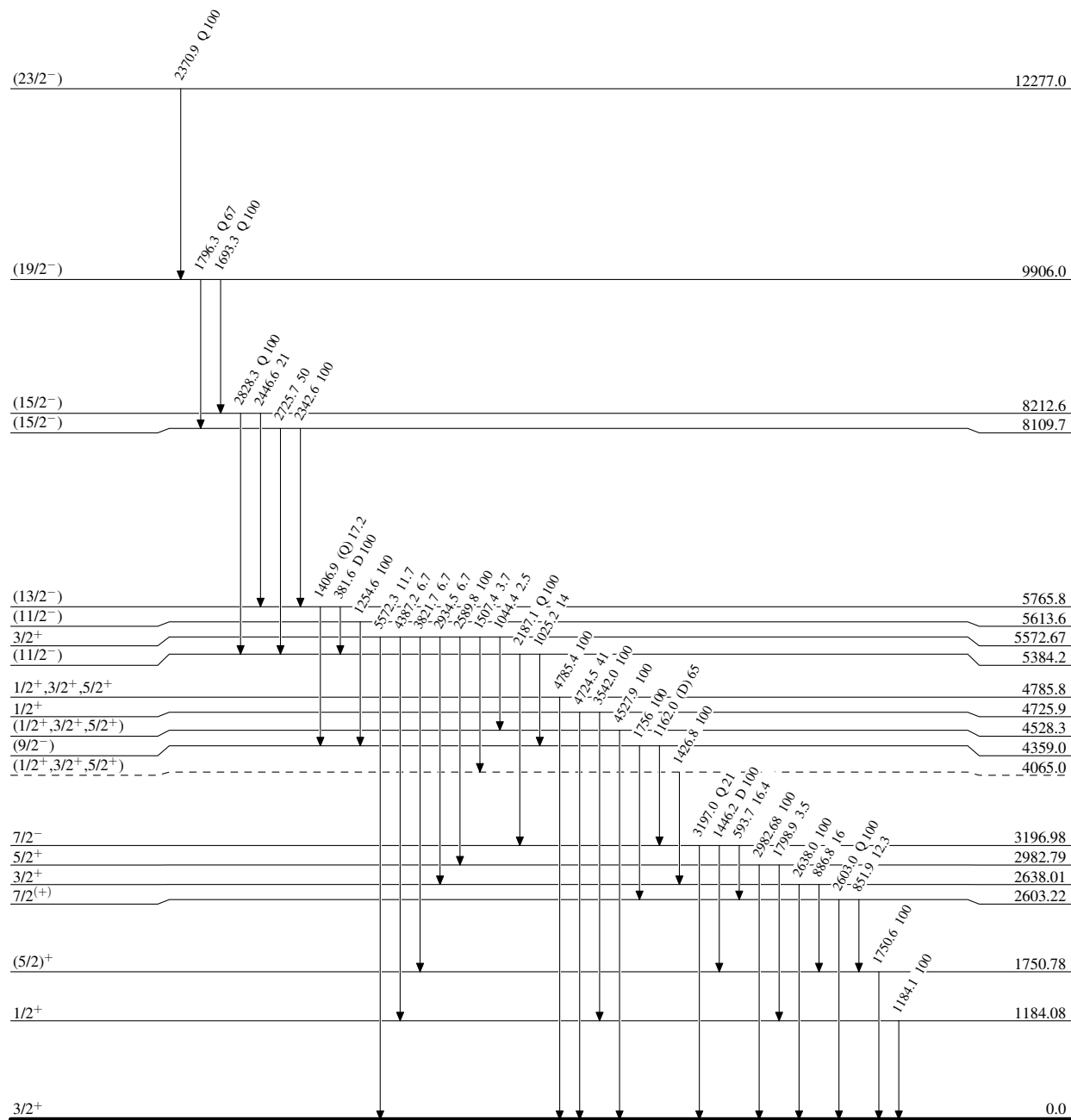
 $\gamma(^{35}\text{Ar})$ (continued)

[‡] Deduced by evaluators from measured $\gamma\gamma(\theta)$ (ADO) in $(^{16}\text{O},\alpha n\gamma)$ and ratios of yields $R(\gamma(\theta))$ in $(^{24}\text{Mg},\alpha n\gamma)$, unless otherwise noted.

[#] From ^{35}K ε decay.

Adopted Levels, GammasLevel Scheme

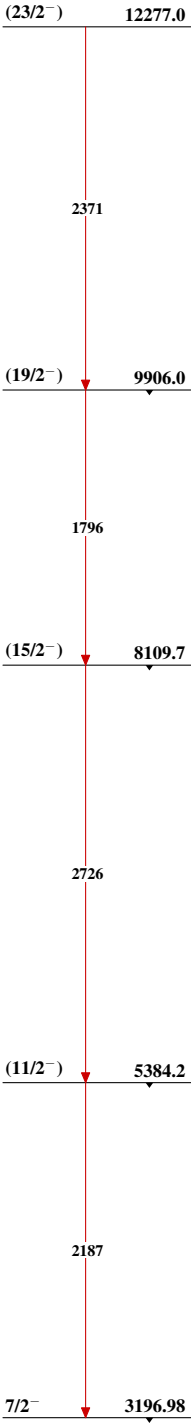
Intensities: Relative photon branching from each level



1.7755 s 14

Adopted Levels, Gammas

Band(A): Band based on f_{7/2}
orbital



³⁵Ar₁₇